

3.3.3 Halogenoalkanes

Halogenoalkanes are much more reactive than alkanes. They have many uses, including as refrigerants, as solvents and in pharmaceuticals. The use of some halogenoalkanes has been restricted due to the effect of chlorofluorocarbons (CFCs) on the atmosphere.

3.3.3.1 Nucleophilic substitution

Content	Opportunities for skills development
Halogenoalkanes contain polar bonds. Halogenoalkanes undergo substitution reactions with the nucleophiles OH^-, CN^- and NH_3 Students should be able to: - outline the nucleophilic substitution mechanisms of these reactions - explain why the carbon–halogen bond enthalpy influences the rate of reaction.	AT a, b and k PS 4.1 Students could follow instructions when carrying out test-tube hydrolysis of halogenoalkanes to show their relative rates of reaction. AT d, g and k Students could prepare a chloroalkane, purifying the product using a separating funnel and distillation.

3.3.3.2 Elimination

Content	Opportunities for skills development
The concurrent substitution and elimination reactions of a halogenoalkane (eg 2-bromopropane with potassium hydroxide). Students should be able to: - explain the role of the reagent as both nucleophile and base - outline the mechanisms of these reactions.	

3.3.3.3 Ozone depletion

Content	Opportunities for skills development
Ozone, formed naturally in the upper atmosphere, is beneficial because it absorbs ultraviolet radiation. Chlorine atoms are formed in the upper atmosphere when ultraviolet radiation causes C–Cl bonds in chlorofluorocarbons (CFCs) to break. Chlorine atoms catalyse the decomposition of ozone and contribute to the hole in the ozone layer. Appreciate that results of research by different groups in the scientific community provided evidence for legislation to ban the use of CFCs as solvents and refrigerants. Chemists have now developed alternative chlorine-free compounds. Students should be able to: • use equations, such as the following, to explain how chlorine atoms catalyse decomposition of ozone: $\text{Cl}\cdot + \text{O}_3 \rightarrow \text{ClO}\cdot + \text{O}_2 \text{ and } \text{ClO}\cdot + \text{O}_3 \rightarrow 2\text{O}_2 + \text{Cl}\cdot$	Research opportunity Students could investigate the role of chemists in the introduction of legislation to ban the use of CFCs and in finding replacements.